

Closing the Lagrange–Legendre–Rvachev Loop

Finite shifted-up synthesis, deconvolved Christoffel–Darboux kernels,
exact oblique projectors, and a rigorous non-flattening theorem

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Abstract

Let $u = u_p$ denote Rvachev’s even C^∞ probability density supported on $[-1, 1]$. The repository already contains two ingredients that at first look like opposite transformations: exact finite synthesis of every polynomial on a compact interval from shifted or shifted-and-dilated copies of u , and an absolutely and uniformly convergent Fourier–Legendre series $u = \sum_{n \geq 0} u_n P_{2n}$. The present report makes the missing basis change explicit and closes the loop.

For arbitrary distinct interpolation nodes $\Xi = \{\xi_0, \dots, \xi_d\} \subset [-1, 1]$, with Lagrange cardinal polynomials ℓ_j , put $M = 2^d$ and let $\mathcal{D}_u = \mathbf{M}_u(D)^{-1}$ be moment deconvolution. We prove the literal-shift formula

$$\ell_j(x) = \frac{1}{M} \sum_{|k| < 2M} (\mathcal{D}_u \ell_j)(k/M) u(x - k/M), \quad -1 \leq x \leq 1.$$

The coefficient matrix B is an exact right inverse of the up-shift collocation matrix U : $UB = I$. This produces an oblique projector BU , a Vandermonde–Appell transmutation, and a basis-independent factorization $B = DV^{-1}$. At Gauss–Legendre nodes it becomes a deconvolved Christoffel–Darboux formula

$$B_{kj} = \frac{\omega_j}{M} K_d^\sharp(k/M, \xi_j), \quad K_d^\sharp(t, y) = \sum_{n=0}^d \frac{2n+1}{2} Q_n(t) P_n(y),$$

where $Q_n = \mathcal{D}_u P_n$. A weighted singular-value identity isolates all conditioning in the deconvolved Legendre block. In the repository’s minimal $d + 2$ -atom endpoint dictionary, the complementary projector is proved to be rank one with an explicit cofactor factorization; functionally, every local up-spline is its nodal Lagrange interpolant plus one scalar multiple of a unique nodal-defect spline.

Substitution into the Legendre expansion closes the loop in blockwise form:

$$u(x) = \sum_{n=0}^{\infty} \frac{u_n}{4^n} \sum_{|k| < 2 \cdot 4^n} Q_{2n}(k/4^n) u(x - k/4^n),$$

with absolute and uniform convergence when each finite inner sum is kept as a block. The report then proves that this block structure is essential. If $C = \mathbf{M}_u(\pi i)$, then

$$Q_{2n}(0) = \frac{2(-1)^n}{C} \frac{(4n)!}{(2n)!(4\pi)^{2n}} \left(1 + \frac{2\pi^2}{4n-1} + \mathcal{O}(n^{-2}) \right),$$

so $|Q_{2n}(0)|^{1/(2n)} \sim 2n/(\pi e)$. Since u is flat and nonanalytic at its endpoints, the central-atom coefficients $u_n Q_{2n}(0)/4^n$ are unbounded in absolute value. Thus grouping the repeated

central copies $u(x)$ is impossible, and the flattened atom series is not absolute or unconditional; small Legendre blocks arise from increasingly violent internal cancellation. Exact symbolic computations, FFT checks through degree 10, and an 80-digit minimal-dictionary certificate accompany the report.

Status and novelty discipline

The Fourier product, moment generating function, exact zero multiplicities, Appell deconvolution, global polynomial synthesis, minimal $d + 2$ endpoint basis, canonical Fourier–Legendre expansion, and reciprocal-MGF pole asymptotics are imported from the current **ProveIt** corpus. The Lagrange specialization itself is a direct but task-specific corollary of the existing synthesis theorem. Claims marked “new here” mean absent from the canonical live exposition and the directly adjacent consolidated frontier volumes after targeted full-text searches; they are corpus-relative claims, not assertions of unqualified worldwide priority. Conjectures and open problems are explicitly labeled. Numerical calculations are supporting checks, never substitutes for proofs.

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1 Orientation, terminology, and corpus audit

1.1 Lagrange is not Legendre

The prompt deliberately puts two easily confused names in one sentence. Throughout this report:

- *Lagrange polynomials* are the cardinal interpolation polynomials ℓ_j associated with arbitrary distinct nodes $\xi_j \in [-1, 1]$;
- *Legendre polynomials* are the orthogonal polynomials P_n with $P_n(1) = 1$.

The repository’s existing convergent series for u is a Fourier–Legendre series, not a series in Lagrange cardinal functions. The loop therefore has three legs:

nodal data $\longleftrightarrow$ Lagrange cardinals $\longleftrightarrow$ Legendre coordinates $\longleftrightarrow$ finite up-atom coefficients.

Gauss–Legendre nodes are the natural point at which all three coordinate systems meet without approximation.

1.2 What was read and what was used

The audit covered the canonical primary TeX exposition, the current semi-formalized frontier volume and its recursive provenance inventory, and the directly adjacent consolidated reports *Exact Polynomial Synthesis from Rvachev Up-Atoms* and *Inverse and Sampling Frontiers for the Fabius–Rvachev System*. Older byte-identical or superseded drafts were treated through the corpus’s own provenance hashes and merge tables rather than counted as mathematically independent sources. Targeted searches were made for “Lagrange”, “Legendre”, “Christoffel–Darboux”, “right inverse”, “collocation”, “cofactor”, and combinations with “up” and “Rvachev”.

The following table separates imported foundations from the task-specific and new deductions.

Status	Result used in this report
Imported	$\widehat{u}(\xi) = \prod_{j \geq 0} \text{sinc}_\pi(\xi/2^j)$, $\text{ord}_{\xi=m} \widehat{u} = 1 + \nu_2(m)$, the probability/MGF model, and the Bernoulli cumulants.
Imported	Exact polynomial synthesis by $M = 2^d$ up-atoms, its physical-scale interval form, the complete dyadic alias nullspace, and the unique $d + 2$ endpoint basis.
Imported	The exact rational Fourier–Legendre coefficients u_n , absolute and uniform convergence of $\sum u_n P_{2n}$, and least-squares optimality of its partial sums.
Imported	The reciprocal-MGF three-shell asymptotic, including the dominant poles $\pm 2\pi i$ and the constant $C = \mathbf{M}_u(\pi i)$.
Task-specific corollary	The explicit finite literal-shift formula for every Lagrange cardinal ℓ_j on $[-1, 1]$, including Appell/Bell–Bernoulli coefficients.
New here, corpus-relative	The matrix closure $UB = I$, the oblique projector BU , moment transmutation, general basis triangle, and Cauchy–Binet identities.
New here, corpus-relative	The deconvolved Christoffel–Darboux kernel, Gauss-cardinal coefficient formula, and weighted singular-value equivalence.
New here, corpus-relative	The cofactor factorization $I - \widehat{B}\widehat{U} = v\lambda^T$ in the minimal $d + 2$ dictionary and its one-scalar nodal-defect decomposition.
New here, corpus-relative	The exact blockwise Legendre–up self-loop, fixed-scale and linearly compressed partial loops, the asymptotic of $Q_{2n}(0)$, and the rigorous non-flattening theorem.

Table 1: Audit status of the principal ingredients.

2 Imported foundations and notation

2.1 The up density, Fourier product, and moment generating function

Let $u = \text{up}$ be the even nonnegative C^∞ function supported on $[-1, 1]$ and normalized by $\int_{-1}^1 u(x) dx = 1$. We use the repository’s Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i \xi x} dx, \quad \widehat{u}(\xi) = \prod_{j=0}^{\infty} \text{sinc}_\pi(\xi/2^j), \quad \text{sinc}_\pi(t) = \frac{\sin(\pi t)}{\pi t}.$$

At every nonzero integer m ,

$$\text{ord}_{\xi=m} \widehat{u}(\xi) = 1 + \nu_2(m).$$

Equivalently, if V_j are independent uniform random variables on $[-1, 1]$, then

$$Z = \sum_{j \geq 1} 2^{-j} V_j$$

has density u . Its moment generating function is

$$\mathbf{M}_u(z) = \mathbb{E} e^{zZ} = \int_{-1}^1 e^{zx} u(x) dx = \prod_{j \geq 1} \frac{\sinh(z/2^j)}{z/2^j}. \quad (1)$$

Write

$$\log \mathbf{M}_u(z) = \sum_{r \geq 1} \kappa_r \frac{z^r}{r!}, \quad \kappa_{2r} = \frac{B_{2r}}{2r(1 - 4^{-r})}, \quad \kappa_{2r+1} = 0. \quad (2)$$

The Bernoulli numbers therefore control every polynomial coefficient appearing below.

2.2 Reciprocal MGF and the Rvachev–Appell family

Define reciprocal coefficients γ_r by

$$\frac{1}{M_u(z)} = \sum_{r=0}^{\infty} \gamma_r \frac{z^r}{r!}. \quad (3)$$

All odd γ_r vanish. On polynomials the formal differential operator

$$\mathcal{D}_u := M_u(D)^{-1} = \sum_{r \geq 0} \gamma_r \frac{D^r}{r!} \quad (4)$$

terminates exactly. Its Appell sequence is

$$A_n(x) := \mathcal{D}_u x^n, \quad \frac{e^{xz}}{M_u(z)} = \sum_{n=0}^{\infty} A_n(x) \frac{z^n}{n!}. \quad (5)$$

In complete Bell-polynomial notation,

$$\gamma_n = Y_n(0, -\kappa_2, 0, -\kappa_4, \dots, -\kappa_n), \quad A_n(x) = Y_n(x, -\kappa_2, 0, -\kappa_4, \dots, -\kappa_n). \quad (6)$$

Thus every coefficient in this report is a finite Bell polynomial in Bernoulli numbers.

$$\begin{aligned} A_0(x) &= 1 \\ A_1(x) &= x \\ A_2(x) &= x^2 - \frac{1}{9} \\ A_3(x) &= x^3 - \frac{x}{3} \\ A_4(x) &= x^4 - \frac{2x^2}{3} + \frac{31}{675} \\ A_5(x) &= x^5 - \frac{10x^3}{9} + \frac{31x}{135} \end{aligned}$$

$$\begin{aligned} Q_0(x) &= 1 \\ Q_1(x) &= x \\ Q_2(x) &= \frac{3x^2}{2} - \frac{2}{3} \\ Q_3(x) &= \frac{5x^3}{2} - \frac{7x}{3} \\ Q_4(x) &= \frac{35x^4}{8} - \frac{20x^2}{3} + \frac{134}{135} \\ Q_5(x) &= \frac{63x^5}{8} - \frac{35x^3}{2} + \frac{33x}{5} \end{aligned}$$

2.3 Exact polynomial synthesis already available in the corpus

The basic imported theorem is the following.

Theorem 2.1 (Global exact synthesis; imported). *Let P have degree at most d , let $M = 2^d$, and put*

$$Q = M_u(MD)^{-1}P.$$

Then, globally on $\mathbb{R}$,

$$P(t) = \frac{1}{M} \sum_{k \in \mathbb{Z}} Q(k) u\left(\frac{t-k}{M}\right), \quad (7)$$

where the sum is locally finite.

The physical-scale form will be more convenient for literal, unscaled translates on $[-1, 1]$.

Theorem 2.2 (Physical-scale interval synthesis; imported). *Let $p \in \Pi_d$, let $I = [A, B]$, choose a spacing $h > 0$, an origin x_0 , and an integer ratio m satisfying $\nu_2(m) \geq d$. Put $\rho = mh$ and $p_\rho^\# = M_u(\rho D)^{-1}p$. Then*

$$p(x) = \frac{1}{m} \sum_{k=\lceil (A-x_0)/h-m \rceil}^{\lfloor (B-x_0)/h+m \rfloor} p_\rho^\#(x_0 + kh) u\left(\frac{x-x_0-kh}{\rho}\right), \quad x \in [A, B]. \quad (8)$$

The proof in the corpus is a Poisson-summation collapse: the exact integer-zero multiplicities annihilate all aliases through degree d , while $M_u(\rho D)^{-1}$ reverses the remaining zero-frequency smoothing.

3 Finite literal-shift synthesis of Lagrange cardinal polynomials

3.1 The main cardinal formula

Fix pairwise distinct nodes

$$\Xi = \{\xi_0, \dots, \xi_d\} \subset [-1, 1]$$

and define the Lagrange cardinal polynomials

$$\ell_j(x) = \prod_{\substack{0 \leq r \leq d \\ r \neq j}} \frac{x - \xi_r}{\xi_j - \xi_r}, \quad \ell_j(\xi_i) = \delta_{ij}. \quad (9)$$

Theorem 3.1 (Literal shifted-up representation of Lagrange cardinals). *Let $M = 2^d$. For every $0 \leq j \leq d$ and every $x \in [-1, 1]$,*

$$\ell_j(x) = \frac{1}{M} \sum_{|k| < 2M} (\mathcal{D}_u \ell_j)(k/M) u(x - k/M). \quad (10)$$

All atoms are literal translates of the same unscaled function u ; only their centers k/M and scalar coefficients vary.

Proof. Apply [theorem 2.2](#) with $A = -1$, $B = 1$, $x_0 = 0$, $m = M$, $h = M^{-1}$, and hence $\rho = 1$. The deconvolved polynomial is $\mathcal{D}_u \ell_j$. The formal limits include $k = \pm 2M$, but for $x \in [-1, 1]$ those terms evaluate u only at points outside $(-1, 1)$, including the zero endpoints, so they vanish identically. Removing them gives [\(10\)](#). $\square$

Remark 3.2 (Atom count versus data complexity). The formula displays $4M - 1 = 2^{d+2} - 1$ atoms, but the coefficient vector is not independent data: it is obtained by sampling one degree- d polynomial $\mathcal{D}_u \ell_j$. Thus its exact description still requires only $\mathcal{O}(d)$ field elements. The later endpoint compression reduces the displayed atom count to $d + 2$ at the cost of a very large common dilation.

3.2 Closed coefficients through symmetric functions and Appell polynomials

Put

$$\lambda_j = \left(\prod_{r \neq j} (\xi_j - \xi_r) \right)^{-1},$$

and let $e_s^{(j)}$ be the elementary symmetric polynomial of degree s in the d nodes $\{\xi_r : r \neq j\}$. Then

$$\ell_j(x) = \lambda_j \sum_{n=0}^d (-1)^{d-n} e_{d-n}^{(j)} x^n.$$

Proposition 3.3 (Appell form of every cardinal coefficient). *For all $t \in \mathbb{R}$,*

$$(\mathcal{D}_u \ell_j)(t) = \sum_{r=0}^{\lfloor d/2 \rfloor} \gamma_{2r} \frac{\ell_j^{(2r)}(t)}{(2r)!} \quad (11)$$

$$= \lambda_j \sum_{n=0}^d (-1)^{d-n} e_{d-n}^{(j)} A_n(t). \quad (12)$$

Consequently, every coefficient in (10) is an explicit rational combination of complete Bell polynomials in the Bernoulli cumulants whenever the nodes are rational.

Proof. Equation (11) is the terminating operator (4); odd terms vanish. Apply $\mathcal{D}_u x^n = A_n(x)$ to the power expansion of ℓ_j to obtain (12). $\square$

3.3 A complete degree-two example

For nodes $(-1, 0, 1)$,

$$\ell_-(x) = \frac{x^2 - x}{2}, \quad \ell_0(x) = 1 - x^2, \quad \ell_+(x) = \frac{x^2 + x}{2}.$$

Because $\gamma_2 = -1/9$,

$$\begin{aligned} \mathcal{D}_u \ell_-(x) &= \frac{x^2 - x - 1/9}{2}, \\ \mathcal{D}_u \ell_0(x) &= \frac{10}{9} - x^2, \\ \mathcal{D}_u \ell_+(x) &= \frac{x^2 + x - 1/9}{2}. \end{aligned}$$

With $M = 4$, each cardinal is therefore a sum over the fifteen centers $k/4$, $-7 \leq k \leq 7$. No numerical linear system is being solved: all coefficients are exact rational values of the displayed quadratics.

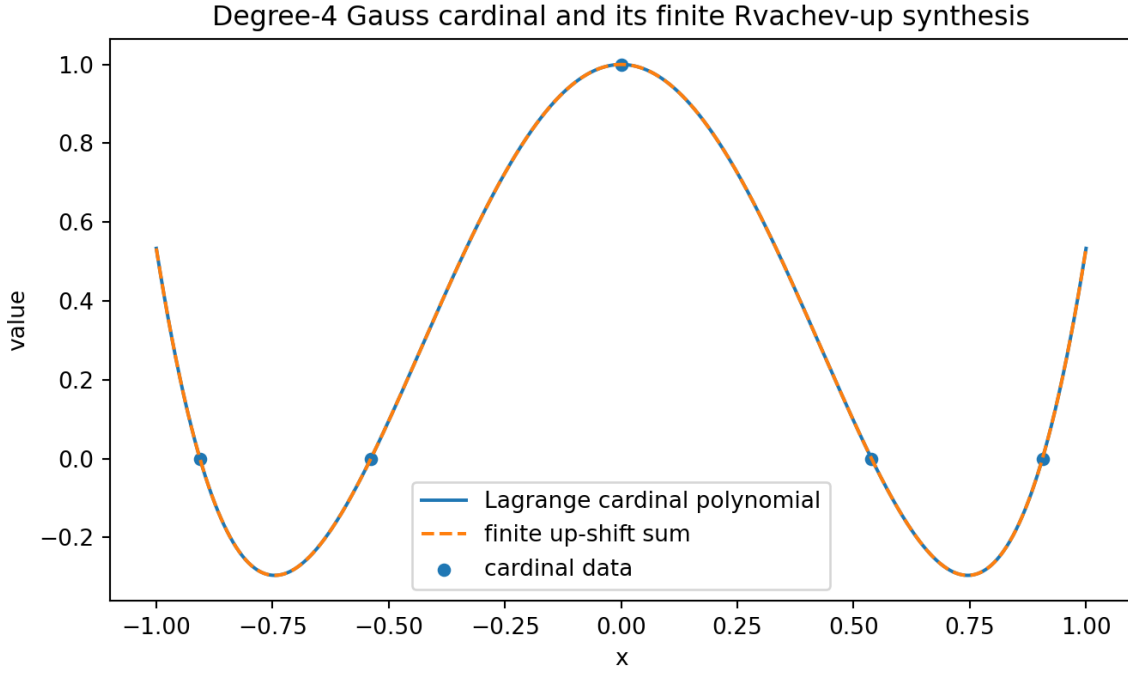


Figure 1: A degree-four Gauss–Legendre cardinal polynomial and its finite literal-shift up synthesis. The two curves are theoretically identical; the visible dashed overlay is a floating-point FFT evaluation.

4 The exact matrix loop: right inverses, projectors, and moments

4.1 Collocation and synthesis matrices

Let

$$\mathcal{K}_d = \{k \in \mathbb{Z} : |k| < 2M\}, \quad q_d = |\mathcal{K}_d| = 4M - 1.$$

Define

$$U_{ik} = u(\xi_i - k/M), \quad B_{kj} = \frac{1}{M}(\mathcal{D}_u \ell_j)(k/M), \quad (13)$$

where rows of U are indexed by nodes and columns by $k \in \mathcal{K}_d$. Thus $U \in \mathbb{R}^{(d+1) \times q_d}$ and $B \in \mathbb{R}^{q_d \times (d+1)}$.

Theorem 4.1 (Exact nodal closure and oblique projector; new here). *The matrices in (13) satisfy*

$$\boxed{UB = I_{d+1}}. \quad (14)$$

Consequently

$$\mathcal{P} := BU, \quad \mathcal{P}^2 = \mathcal{P}, \quad \text{rank } \mathcal{P} = \text{tr } \mathcal{P} = d + 1, \quad (15)$$

with

$$\text{range } \mathcal{P} = \text{range } B, \quad \ker \mathcal{P} = \ker U. \quad (16)$$

In general $\mathcal{P}$ is oblique rather than orthogonal.

Proof. Evaluate (10) at $x = \xi_i$. The result is $(UB)_{ij} = \ell_j(\xi_i) = \delta_{ij}$. All remaining identities follow from $UB = I$: $\mathcal{P}^2 = B(UB)U = BU$, the range and kernel assertions are standard, and the nonzero eigenvalues of an idempotent are all 1. $\square$

This matrix identity gives the precise finite-dimensional meaning of “closing the Lagrange loop”:

$$\boxed{\text{nodal values} \xrightarrow{B} \text{up-shift coefficients} \xrightarrow{U} \text{the same nodal values.}}$$

4.2 Partition, Appell moments, and Cauchy–Binet identities

Let $\mathbf{1}$ denote an all-ones vector and write $\boldsymbol{\xi}^{\text{or}} = (\xi_0^r, \dots, \xi_d^r)^T$.

Proposition 4.2 (Vandermonde–Appell transmutation; new here). *For $0 \leq r \leq d$,*

$$B\mathbf{1} = \frac{1}{M}\mathbf{1}, \quad B\boldsymbol{\xi}^{\text{or}} = \frac{1}{M}(A_r(k/M))_{k \in \mathcal{K}_d}. \quad (17)$$

Moreover,

$$1 = \det(UB) = \sum_{\substack{S \subset \mathcal{K}_d \\ |S|=d+1}} \det U_{:,S} \det B_{S,:}. \quad (18)$$

Proof. The nodal vector $\boldsymbol{\xi}^{\text{or}}$ interpolates the polynomial x^r exactly. Its deconvolution is A_r , so (17) follows from [theorem 3.1](#). The first identity is the case $r = 0$. Equation (18) is the Cauchy–Binet formula applied to $UB = I$. $\square$

Equation (18) is a nontrivial determinant partition of unity over all square subdictionaries. It can be used to compare sparse choices of centers without recomputing the interpolation problem itself.

4.3 Any polynomial basis fits the same triangle

Let $p_0, \dots, p_d$ be any basis of Π_d and define

$$V_{in} = p_n(\xi_i), \quad D_{kn} = \frac{1}{M}(\mathcal{D}_u p_n)(k/M). \quad (19)$$

Theorem 4.3 (Basis-independent factorization; new here). *With (19),*

$$UD = V, \quad B = DV^{-1}, \quad D = BV. \quad (20)$$

Thus the Lagrange-to-up transform is exactly the composition of a standard basis change V^{-1} and basis-polynomial deconvolution/sampling D .

Proof. The n th column of UD is the vector obtained by evaluating the exact up synthesis of p_n at all nodes, hence it equals the n th column of V . The columns of V^{-1} are the coordinates of the cardinal polynomials in the basis p_n , so applying D gives $B = DV^{-1}$. Multiplication by V gives the last identity. $\square$

Algorithm 4.4 (Exact cardinal synthesis without expanding every cardinal). Given nodes ξ_i and a numerically or symbolically convenient polynomial basis p_n :

1. form the small $(d+1) \times (d+1)$ Vandermonde-type matrix V ;
2. compute the $q_d \times (d+1)$ table $D_{kn} = M^{-1}(\mathcal{D}_u p_n)(k/M)$;
3. output $B = DV^{-1}$.

For Gauss–Legendre nodes, [section 5](#) removes the general matrix inverse entirely.

5 Gauss–Legendre nodes and a deconvolved Christoffel–Darboux kernel

5.1 Deconvolved Legendre polynomials

Set

$$Q_n(x) := \mathcal{D}_u P_n(x). \quad (21)$$

Using the standard power expansion of P_n gives an explicit Appell formula.

Proposition 5.1 (Appell expansion and parity). *For every $n \geq 0$,*

$$Q_n(x) = 2^{-n} \sum_{r=0}^{\lfloor n/2 \rfloor} (-1)^r \frac{(2n-2r)!}{r!(n-r)!(n-2r)!} A_{n-2r}(x). \quad (22)$$

In particular, $Q_n(-x) = (-1)^n Q_n(x)$.

Proof. Apply $\mathcal{D}_u x^m = A_m(x)$ termwise to

$$P_n(x) = 2^{-n} \sum_{r=0}^{\lfloor n/2 \rfloor} (-1)^r \frac{(2n-2r)!}{r!(n-r)!(n-2r)!} x^{n-2r}.$$

Parity follows because $\mathcal{D}_u$ contains only even derivatives. $\square$

The first few Q_n were displayed in [section 2](#). Unlike the Legendre family, they are not orthogonal; they are the exact coefficient polynomials that undo convolution by the up law.

5.2 A nonlocal three-term recurrence

Define the odd differential symbol

$$\Lambda(z) := -\frac{M'_u(z)}{M_u(z)} = -\sum_{r \geq 1} \kappa_{2r} \frac{z^{2r-1}}{(2r-1)!}. \quad (23)$$

Proposition 5.2 (Deconvolved Legendre recurrence; new here). *For $n \geq 1$,*

$$\boxed{(n+1)Q_{n+1} = (2n+1)[xQ_n + \Lambda(D)Q_n] - nQ_{n-1}.} \quad (24)$$

Equivalently,

$$(n+1)Q_{n+1} = (2n+1) \left[xQ_n - \sum_{r \geq 1} \kappa_{2r} \frac{Q_n^{(2r-1)}}{(2r-1)!} \right] - nQ_{n-1}, \quad (25)$$

where the derivative sum terminates.

Proof. For any formal power series g , the operator commutator is $g(D)x = xg(D) + g'(D)$. Taking $g = 1/M_u$ gives

$$\mathcal{D}_u(xf) = x\mathcal{D}_u f + \frac{g'(D)}{g(D)} \mathcal{D}_u f = x\mathcal{D}_u f + \Lambda(D)\mathcal{D}_u f.$$

Apply $\mathcal{D}_u$ to the classical Legendre recurrence $(n+1)P_{n+1} = (2n+1)xP_n - nP_{n-1}$. $\square$

5.3 The kernel theorem

Let $\xi_0, \dots, \xi_d$ now be the zeros of P_{d+1} and let $\omega_0, \dots, \omega_d$ be the Gauss–Legendre weights. Write

$$h_n = \int_{-1}^1 P_n(x)^2 dx = \frac{2}{2n+1}, \quad K_d(x, y) = \sum_{n=0}^d \frac{P_n(x)P_n(y)}{h_n}. \quad (26)$$

Define its deconvolution in the first variable by

$$K_d^\sharp(t, y) := \mathcal{D}_{u,t} K_d(t, y) = \sum_{n=0}^d \frac{Q_n(t)P_n(y)}{h_n}. \quad (27)$$

Theorem 5.3 (Deconvolved Christoffel–Darboux synthesis; new here). *Let $M = 2^d$. For $x, y \in [-1, 1]$,*

$$K_d(x, y) = \frac{1}{M} \sum_{|k| < 2M} K_d^\sharp(k/M, y) u(x - k/M). \quad (28)$$

The Gauss cardinal polynomials satisfy

$$\ell_j(x) = \omega_j K_d(x, \xi_j), \quad (29)$$

and therefore their up coefficients are

$$B_{kj} = \frac{\omega_j}{M} K_d^\sharp(k/M, \xi_j). \quad (30)$$

Equivalently, the following discrete biorthogonality holds:

$$\frac{\omega_j}{M} \sum_{|k| < 2M} K_d^\sharp(k/M, \xi_j) u(\xi_i - k/M) = \delta_{ij}. \quad (31)$$

Proof. For fixed y , the function $x \mapsto K_d(x, y)$ is a polynomial of degree at most d . Its deconvolution is exactly $K_d^\sharp(\cdot, y)$, so [theorem 3.1](#) applied to this polynomial gives (28). Gauss quadrature is exact on products of two polynomials of degree at most d , hence for $p \in \Pi_d$,

$$p(x) = \int_{-1}^1 K_d(x, y) p(y) dy = \sum_{j=0}^d \omega_j K_d(x, \xi_j) p(\xi_j).$$

Uniqueness of cardinal interpolation proves (29). Substitute (28) and compare with (13); then evaluate at $x = \xi_i$ to obtain (31). $\square$

5.4 A weighted singular-value identity

Let

$$V_{in} = P_n(\xi_i), \quad W = \text{diag}(\omega_0, \dots, \omega_d), \quad H = \text{diag}(h_0, \dots, h_d),$$

and let $D_{kn} = M^{-1} Q_n(k/M)$ as in [theorem 4.3](#). Discrete orthogonality gives $V^T W V = H$.

Theorem 5.4 (Gauss transform is condition-neutral in natural weights; new here). *The matrix*

$$O = W^{1/2}VH^{-1/2}$$

is orthogonal, and

$$BW^{-1/2} = DH^{-1/2}O^T. \quad (32)$$

Therefore

$$\boxed{\sigma_j(BW^{-1/2}) = \sigma_j(DH^{-1/2}) \quad (0 \leq j \leq d),} \quad (33)$$

including equality of spectral norms and condition numbers on the nonzero singular spectrum.

Proof. The identity $V^TWV = H$ gives $O^TO = I$; since O is square, it is orthogonal. Solving $V = W^{-1/2}OH^{1/2}$ gives $V^{-1} = H^{-1/2}O^TW^{1/2}$. Insert this into $B = DV^{-1}$ and right-multiply by $W^{-1/2}$. Right multiplication by an orthogonal matrix preserves all singular values. $\square$

This is a useful diagnostic: at Gauss nodes, poor conditioning is not created by the Lagrange/Legendre transform itself. It is already present in the sampled deconvolution polynomials Q_n .

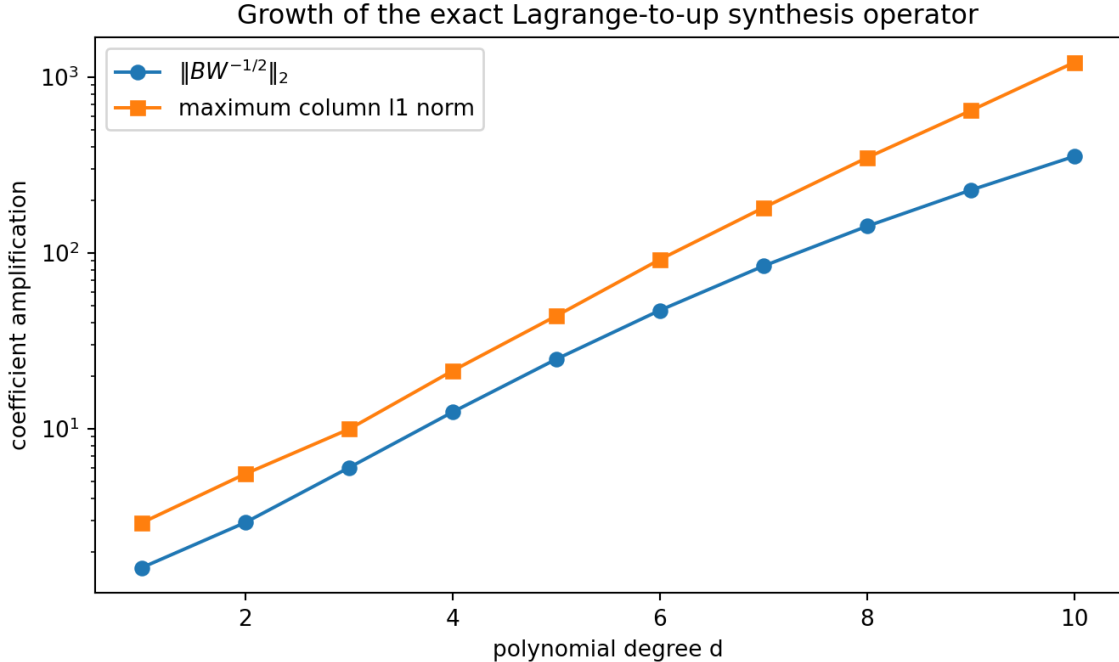


Figure 2: Double-precision growth of the weighted synthesis norm and the largest column ℓ^1 norm for degrees 1 through 10. The exact weighted singular-value identity was verified to errors between 2.2×10^{-16} and 1.7×10^{-13} .

6 The minimal $d + 2$ -atom dictionary and its rank-one complement

6.1 Imported endpoint compression, specialized to interpolation

Map $x \in [-1, 1]$ to $t = (x + 1)/2 \in [0, 1]$ and put $M = 2^d$. The imported endpoint theorem retains the $d + 2$ integer centers

$$k_r = M - d - 1 + r, \quad 0 \leq r \leq d + 1, \quad (34)$$

and the normalized atoms

$$\Psi_r(x) = \frac{1}{M} u\left(\frac{x+1-2k_r}{2M}\right). \quad (35)$$

For each cardinal polynomial there is a unique coefficient vector $\hat{B}_{:,j} \in \mathbb{R}^{d+2}$ such that

$$\ell_j(x) = \sum_{r=0}^{d+1} \hat{B}_{rj} \Psi_r(x), \quad -1 \leq x \leq 1. \quad (36)$$

The coefficients are produced exactly by the corpus's dyadic annihilator

$$A_d(z) = \prod_{m=1}^d [2^m]_z, \quad [N]_z = 1 + z + \cdots + z^{N-1}, \quad (37)$$

whose degree is $2M - d - 2$. This is where the q -integer and Thue–Morse structures enter the compression.

Let

$$\hat{U}_{ir} = \Psi_r(\xi_i). \quad (38)$$

Then evaluating (36) at the nodes gives

$$\hat{U}\hat{B} = I_{d+1}. \quad (39)$$

Here $\hat{U}$ is $(d+1) \times (d+2)$ and $\hat{B}$ is $(d+2) \times (d+1)$, so the coefficient space has exactly one redundant direction.

6.2 Cofactors identify the unique redundant direction

For $0 \leq r \leq d+1$, let $\hat{U}_{\hat{r}}$ denote the square matrix obtained by deleting column r of $\hat{U}$, and let $\hat{B}_{\hat{r}}$ denote the square matrix obtained by deleting row r of $\hat{B}$. Define cofactor vectors

$$v_r = (-1)^r \det(\hat{U}_{\hat{r}}), \quad \lambda_r = (-1)^r \det(\hat{B}_{\hat{r}}). \quad (40)$$

Theorem 6.1 (Rank-one complement of the minimal interpolation loop; new here). *The vectors in (40) satisfy*

$$\hat{U}v = 0, \quad \lambda^T \hat{B} = 0, \quad \lambda^T v = 1. \quad (41)$$

Moreover,

$$\boxed{I_{d+2} - \hat{B}\hat{U} = v\lambda^T.} \quad (42)$$

Thus $\lambda^T a = 0$ is the unique scalar polynomiality certificate for a minimal-dictionary coefficient vector a , and v is the unique coefficient direction whose up-spline vanishes at every interpolation node.

Proof. The alternating maximal minors of a full-row-rank matrix give a right-null vector, so $\hat{U}v = 0$. The same cofactor argument applied to $\hat{B}^T$ gives $\lambda^T \hat{B} = 0$. Cauchy–Binet and $\hat{U}\hat{B} = I$ yield

$$1 = \det(\hat{U}\hat{B}) = \sum_{r=0}^{d+1} \det(\hat{U}_{\hat{r}}) \det(\hat{B}_{\hat{r}}) = \lambda^T v.$$

The idempotent $\hat{B}\hat{U}$ has rank $d+1$, so its complementary idempotent has rank one, range $\ker \hat{U} = \text{span}\{v\}$, and left range $\ker \hat{B}^T = \text{span}\{\lambda\}$. The normalization $\lambda^T v = 1$ determines the factorization uniquely. $\square$

Corollary 6.2 (Every local up-spline is a polynomial plus one nodal defect). *For $a \in \mathbb{R}^{d+2}$ define*

$$F_a(x) = \sum_{r=0}^{d+1} a_r \Psi_r(x), \quad G_\Xi(x) = \sum_{r=0}^{d+1} v_r \Psi_r(x).$$

Then $G_\Xi(\xi_i) = 0$ for all i , and

$$\boxed{F_a(x) = I_\Xi F_a(x) + (\lambda^T a) G_\Xi(x)}, \quad (43)$$

where $I_\Xi F_a$ is the degree- d Lagrange interpolant of the nodal values of F_a . In particular, $F_a \in \Pi_d$ on $[-1, 1]$ if and only if $\lambda^T a = 0$.

Proof. The coefficient vector of $I_\Xi F_a$ in the minimal dictionary is $\widehat{B}\widehat{U}a$. Subtract this from a and use (42). The endpoint dictionary spans a space of dimension $d + 2$ whose polynomial subspace has dimension $d + 1$, so the final criterion is exact. $\square$

This result is the nodal-coordinate form of the one-dimensional quotient found in the repository's local dimension theorem. Any previously chosen Thue–Morse defect functional on that quotient must be proportional to λ^T ; [theorem 6.1](#) fixes the proportionality by cofactors.

6.3 Readable and high-cancellation examples

For $d = 2$ and nodes $(-1, 0, 1)$, the retained centers are $k = (1, 2, 3, 4)$ and the exact coefficient matrix is

$$\widehat{B}_2 = \begin{bmatrix} \frac{392}{9} & -\frac{496}{9} & \frac{176}{9} \\ -\frac{752}{680} & \frac{1072}{1072} & -\frac{392}{464} \\ -\frac{464}{9} & \frac{496}{9} & -\frac{104}{9} \end{bmatrix}, \quad k = (1, 2, 3, 4).$$

The entries are already much larger than the cardinal polynomial values, foreshadowing cancellation.

For $d = 4$, nodes $(-1, -1/2, 0, 1/2, 1)$, and $M = 16$, the six retained centers are 11, 12, 13, 14, 15, 16. The largest exact coefficient in the generated matrix is

$$\frac{10524639232}{675} \approx 1.5592 \times 10^7.$$

Nevertheless, evaluating $\widehat{U}\widehat{B}$ with 80-digit arithmetic and a 3000-term odd-half-integer cosine series for u gave

$$\max_{i,j} |(\widehat{U}\widehat{B} - I)_{ij}| = 1.1463 \times 10^{-24}.$$

The residual is dominated by truncating the cosine series; the symbolic coefficient construction is exact.

7 The literal Lagrange interpolation loop

7.1 Arbitrary nodal data

For a function f known at the nodes, let

$$I_\Xi f(x) = \sum_{j=0}^d f(\xi_j) \ell_j(x) \quad (44)$$

be its degree- d interpolant.

Theorem 7.1 (Finite up-shift realization of any interpolant). *With $M = 2^d$,*

$$I_{\Xi}f(x) = \frac{1}{M} \sum_{|k| < 2M} c_{d,k}(f) u(x - k/M), \quad (45)$$

where

$$c_{d,k}(f) = \sum_{j=0}^d f(\xi_j) (\mathcal{D}_u \ell_j)(k/M) = (\mathcal{D}_u I_{\Xi}f)(k/M). \quad (46)$$

Proof. Multiply (10) by $f(\xi_j)$, sum over j , and use linearity of $\mathcal{D}_u$. $\square$

For $f = u$, (45) is already a finite self-representation: it writes a polynomial approximation of u as a finite sum of literal translates of u itself.

7.2 Superalgebraic convergence at Chebyshev–Lobatto nodes

Let $I_d^{\text{CL}}u$ use the Chebyshev–Lobatto nodes $\cos(j\pi/d)$. Since $u \in C^\infty[-1, 1]$, Jackson estimates give, for every integer $r \geq 1$,

$$E_d(u) := \inf_{p \in \Pi_d} \|u - p\|_\infty = \mathcal{O}_r(d^{-r}).$$

The Chebyshev–Lobatto Lebesgue constants satisfy $\Lambda_d = \mathcal{O}(\log d)$, so Lebesgue’s lemma yields

$$\|u - I_d^{\text{CL}}u\|_\infty \leq (1 + \Lambda_d) E_d(u) = \mathcal{O}_N(d^{-N}) \quad \text{for every } N. \quad (47)$$

Combining (47) with theorem 7.1 produces a sequence of finite literal-shift sums converging uniformly and faster than any fixed power of d^{-1} to u .

There is no geometric conclusion: u is flat but nonanalytic at ± 1 , so a uniform geometric polynomial rate would contradict classical Bernstein ellipse theory.

7.3 Gauss data factor through a discrete Legendre transform

At Gauss–Legendre nodes, define the discrete Legendre coefficients

$$\tilde{f}_n = \frac{2n+1}{2} \sum_{j=0}^d \omega_j f(\xi_j) P_n(\xi_j). \quad (48)$$

Then $I_{\Xi}f = \sum_{n=0}^d \tilde{f}_n P_n$, and

$$I_{\Xi}f(x) = \frac{1}{M} \sum_{|k| < 2M} \left(\sum_{n=0}^d \tilde{f}_n Q_n(k/M) \right) u(x - k/M). \quad (49)$$

Thus nodal interpolation, orthogonal coordinates, and up synthesis form an exact commutative diagram.

7.4 The corresponding projection-kernel loop

For any integrable f , let $\Pi_d f$ be its degree- d Legendre projection. By [theorem 5.3](#),

$$\Pi_d f(x) = \frac{1}{M} \sum_{|k| < 2M} \left[\int_{-1}^1 K_d^\sharp(k/M, y) f(y) dy \right] u(x - k/M). \quad (50)$$

This is the continuous analogue of (49). The bracketed coefficients are not obtained by fitting an up basis; they are exact Legendre moments passed through $\mathcal{D}_u$.

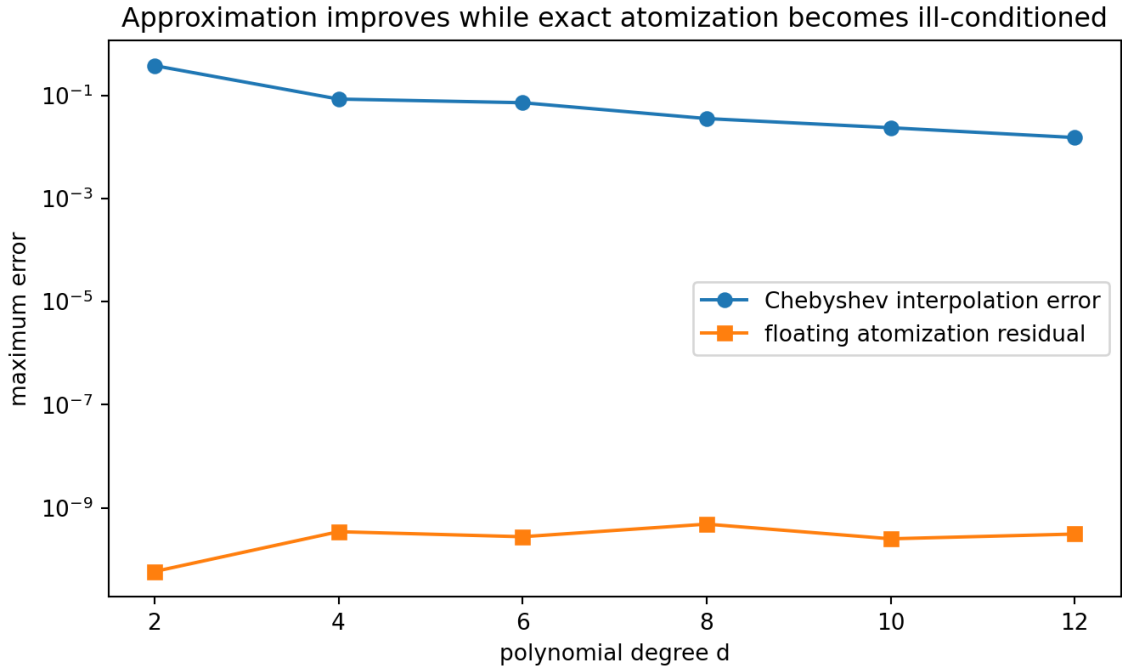


Figure 3: Chebyshev–Lobatto interpolation error for u compared with the floating residual incurred when the exact polynomial interpolant is evaluated through thousands of signed up atoms. Approximation improves while the atomization becomes increasingly cancellation-sensitive.

8 Closing the canonical Fourier–Legendre loop

8.1 The imported series

The canonical repository expansion is

$$u(x) = \sum_{n=0}^{\infty} u_n P_{2n}(x), \quad u_n = \frac{4n+1}{2} \int_{-1}^1 u(x) P_{2n}(x) dx, \quad (51)$$

with absolute and uniform convergence on $[-1, 1]$. Its coefficients have an exact finite formula in dyadic Fabius values; equivalently, they are rational because all moments of the up law are rational.

n	exact u_n	decimal value
0	$\frac{1}{2}$	0.5
1	$-\frac{5}{6}$	-0.833333333333
2	$\frac{11}{30}$	0.366666666667
3	$\frac{143}{5670}$	0.0252204585538
4	$-\frac{7291}{85050}$	-0.0857260435038
5	$\frac{99137}{2485890}$	0.0398798820543

8.2 Termwise finite closure

For the polynomial P_{2n} use degree $d = 2n$, hence $M_n = 2^{2n} = 4^n$. [Theorem 3.1](#) applied to this single polynomial gives

$$P_{2n}(x) = \frac{1}{4^n} \sum_{|k| < 2 \cdot 4^n} Q_{2n}(k/4^n) u(x - k/4^n). \quad (52)$$

Theorem 8.1 (Blockwise Lagrange–Legendre–up closure; new here). *For every $x \in [-1, 1]$,*

$$u(x) = \sum_{n=0}^{\infty} \frac{u_n}{4^n} \sum_{|k| < 2 \cdot 4^n} Q_{2n}(k/4^n) u(x - k/4^n). \quad (53)$$

The outer series converges absolutely and uniformly when each finite inner sum is retained as one block.

Proof. Substitute (52) into (51). The n th finite block is exactly $u_n P_{2n}(x)$. Since $\|P_m\|_{\infty} \leq 1$ on $[-1, 1]$ and the repository proves $\sum_n |u_n| < \infty$,

$$\sum_{n \geq 0} \|u_n P_{2n}\|_{\infty} \leq \sum_{n \geq 0} |u_n| < \infty.$$

□

The theorem is an exact self-representation of u , but it is a series of *finite cancellation blocks*, not yet a single unconditionally convergent series of individual atoms. [Section 9](#) proves that this distinction cannot be removed.

8.3 One fixed lattice for each partial sum

Let

$$S_N(x) = \sum_{n=0}^N u_n P_{2n}(x), \quad C_N(t) = \mathcal{D}_u S_N(t) = \sum_{n=0}^N u_n Q_{2n}(t). \quad (54)$$

The degree of S_N is $2N$, so a single lattice with $M_N = 4^N$ synthesizes the entire partial sum.

Corollary 8.2 (Fixed-scale finite self-approximants). *For $x \in [-1, 1]$,*

$$S_N(x) = \frac{1}{4^N} \sum_{|k| < 2 \cdot 4^N} C_N(k/4^N) u(x - k/4^N). \quad (55)$$

Every displayed coefficient is rational, and the right side converges uniformly to u as $N \rightarrow \infty$.

Proof. Apply the physical-scale synthesis theorem to S_N . Rationality follows from rationality of u_n , of γ_r , and of evaluation of rational polynomials at dyadic rationals. Uniform convergence is exactly the imported uniform convergence $S_N \rightarrow u$. $\square$

Taking $d = 2N$ in the endpoint compression gives an even more economical but more ill-conditioned version.

Corollary 8.3 (Only $2N + 2$ atoms per Legendre partial sum). *Let $M = 4^N$ and put $P_N(t) = S_N(2t - 1)$. There are unique rational coefficients $b_{N,k}$, $M - 2N - 1 \leq k \leq M$, such that*

$$S_N(x) = \frac{1}{M} \sum_{k=M-2N-1}^M b_{N,k} u\left(\frac{x+1-2k}{2M}\right), \quad -1 \leq x \leq 1. \quad (56)$$

The sums in (56) have only $2N + 2$ atoms and converge uniformly to u .

This gives a striking scale–cardinality tradeoff: literal unit-scale shifts need exponentially many displayed centers, while a common dilation of order 4^N permits a linear number of atoms. The latter coefficients exhibit severe cancellation, as [sections 6](#) and [9](#) explain.

8.4 Superalgebraic convergence rate of the closed partial loop

Let

$$\mathcal{L}f = -((1 - x^2)f'(x))'. \quad (57)$$

Then $\mathcal{L}P_m = m(m + 1)P_m$. Boundary terms vanish because of the factor $1 - x^2$, so repeated self-adjointness gives, for every $r \geq 1$,

$$\int_{-1}^1 f P_m = \frac{1}{[m(m + 1)]^r} \int_{-1}^1 (\mathcal{L}^r f) P_m. \quad (58)$$

By Cauchy–Schwarz and $\|P_m\|_2^2 = 2/(2m + 1)$, the normalized Legendre coefficient $a_m = (2m + 1) \int f P_m / 2$ satisfies

$$|a_m| \leq \sqrt{\frac{2m + 1}{2}} \frac{\|\mathcal{L}^r f\|_2}{[m(m + 1)]^r} = \mathcal{O}_r(m^{1/2-2r}). \quad (59)$$

For $f = u$, r is arbitrary. Summing the tail and using $\|P_m\|_\infty \leq 1$ shows

$$\|u - S_N\|_\infty = \mathcal{O}_R(N^{-R}) \quad \text{for every } R > 0. \quad (60)$$

Therefore both exact atomizations (55) and (56) converge superalgebraically as functions, even though their coefficient maps become rapidly unstable.

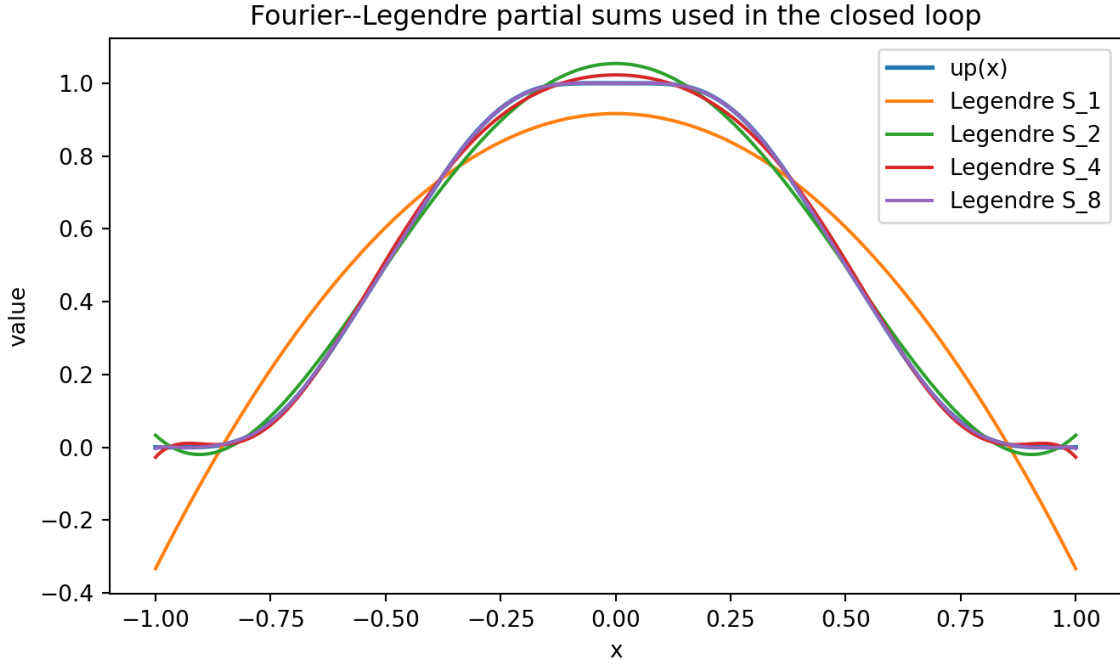


Figure 4: The first Fourier–Legendre partial sums S_N . Each curve also has an exact finite up-atom representation, either on the literal dyadic lattice (55) or in the $2N + 2$ -atom endpoint dictionary (56).

9 Factorial deconvolution and the non-flattening theorem

This section analyzes the closed loop rather than merely writing it down. The main phenomenon is that $\mathcal{D}_u$ is harmless on each fixed polynomial but is a factorially growing transform as the degree increases.

9.1 Reciprocal-MGF pole input

Let

$$C = M_u(\pi i) = \prod_{j \geq 1} \frac{\sin(\pi/2^j)}{\pi/2^j} \approx 0.553771275888 \dots \quad (61)$$

The nearest poles of $1/M_u(z)$ are the simple conjugate pair $\pm 2\pi i$. The repository proves the sharper three-shell formula; in the present notation its leading consequence is

$$\gamma_{2m} = \frac{2(-1)^m (2m)!}{C(2\pi)^{2m}} (1 + \mathcal{O}(m4^{-m})). \quad (62)$$

9.2 A new asymptotic for the deconvolved Legendre constants

Write the even Legendre polynomial as

$$P_{2n}(x) = \sum_{j=0}^n c_{n,j} x^{2n-2j}, \quad c_{n,j} = 2^{-2n} (-1)^j \frac{(4n-2j)!}{j!(2n-j)!(2n-2j)!}. \quad (63)$$

Since $\mathcal{D}_u x^r$ has constant term γ_r ,

$$Q_{2n}(0) = \sum_{j=0}^n c_{n,j} \gamma_{2n-2j}. \quad (64)$$

Theorem 9.1 (Factorial asymptotic of $Q_{2n}(0)$; new here). *As $n \rightarrow \infty$,*

$$Q_{2n}(0) = \frac{2(-1)^n}{C} \frac{(4n)!}{(2n)!(4\pi)^{2n}} \left(1 + \frac{2\pi^2}{4n-1} + \mathcal{O}(n^{-2}) \right). \quad (65)$$

Consequently,

$$Q_{2n}(0) = \frac{2\sqrt{2}}{C} (-1)^n \left(\frac{2n}{\pi e} \right)^{2n} \left(1 + \mathcal{O}(n^{-1}) \right), \quad (66)$$

and

$$|Q_{2n}(0)|^{1/(2n)} \sim \frac{2n}{\pi e}. \quad (67)$$

Proof. Let $T_{n,j} = c_{n,j} \gamma_{2n-2j}$ and first isolate $T_{n,0}$. The coefficient ratio for $j = 1$ is exact:

$$\frac{c_{n,1}}{c_{n,0}} = -\frac{n(2n-1)}{4n-1}. \quad (68)$$

From (62),

$$\frac{\gamma_{2n-2}}{\gamma_{2n}} = -\frac{(2\pi)^2}{(2n)(2n-1)} (1 + \mathcal{O}(n4^{-n})). \quad (69)$$

Multiplication gives

$$\frac{T_{n,1}}{T_{n,0}} = \frac{2\pi^2}{4n-1} + \mathcal{O}(4^{-n}). \quad (70)$$

It remains to control $j \geq 2$ uniformly. The meromorphic asymptotic implies constants $0 < c < C_1$ such that, for all sufficiently large indices,

$$c \frac{(2m)!}{(2\pi)^{2m}} \leq |\gamma_{2m}| \leq C_1 \frac{(2m)!}{(2\pi)^{2m}},$$

with finitely many small indices absorbed into C_1 . Hence

$$\begin{aligned} \left| \frac{T_{n,j}}{T_{n,0}} \right| &\leq C_2 \frac{(4n-2j)!(2n)!}{j!(2n-j)!(4n)!} (2\pi)^{2j} \\ &\leq \frac{C_2}{j!} \left(\frac{2\pi^2}{n} \right)^j, \quad 0 \leq j \leq n. \end{aligned}$$

Indeed, $(2n)!/(2n-j)! \leq (2n)^j$, whereas every factor in $(4n)!/(4n-2j)!$ is at least $2n$. Summing over $j \geq 2$ gives $\mathcal{O}(n^{-2})$. Therefore

$$Q_{2n}(0) = c_{n,0} \gamma_{2n} \left(1 + \frac{2\pi^2}{4n-1} + \mathcal{O}(n^{-2}) \right).$$

Now $c_{n,0} = 2^{-2n} \binom{4n}{2n}$, and substitution of (62) gives (65). Stirling's formula gives (66) and (67). $\square$

Remark 9.2 (Formal degree generating functions have zero radius). Applying $\mathcal{D}_u$ to the classical generating function gives the formal identity

$$\sum_{n \geq 0} Q_n(x) t^n = (1 - 2xt + t^2)^{-1/2} \sum_{r \geq 0} \gamma_r \frac{(1/2)_r}{r!} \left(\frac{2t}{1 - 2xt + t^2} \right)^r. \quad (71)$$

Theorem 9.1 proves that at $x = 0$ this is a purely formal, Gevrey-type expansion with zero ordinary radius of convergence. This is not a contradiction: $\mathcal{D}_u$ was defined as a terminating operator on each fixed polynomial, not as a bounded operator on an analytic function space.

9.3 A new Lambert– W threshold

Let

$$n_Q(X) = \min\{n \geq 1 : |Q_{2n}(0)| \geq X\}.$$

From (66), the leading logarithmic equation is

$$2n \log \frac{2n}{\pi e} = \log X.$$

Corollary 9.3 (Legendre-deconvolution threshold; new here). *As $X \rightarrow \infty$,*

$$n_Q(X) = \frac{\log X}{2W(\log X/(\pi e))} + \mathcal{O}(1). \quad (72)$$

Proof. Set $H = \log X$ and $w = W(H/(\pi e))$. Then $n_0 = H/(2w)$ satisfies $2n_0 \log(2n_0/(\pi e)) = H$. The omitted logarithmic prefactor in (66) changes the continuous solution by $o(1)$, and rounding to the least integer contributes $\mathcal{O}(1)$. $\square$

The repository’s reciprocal-MGF threshold has $W(\log X/(2\pi e))$ instead. The changed argument records the extra central binomial amplification contributed by the leading coefficient of P_{2n} .

9.4 The central atom cannot be grouped

In the n th block of (53), the index $k = 0$ multiplies the same atom $u(x)$ for every n . Its scalar coefficient is

$$\beta_n = \frac{u_n Q_{2n}(0)}{4^n}. \quad (73)$$

Lemma 9.4 (Supergeometric Legendre coefficients force analyticity). *If the even Legendre coefficients of a continuous function f satisfy $|a_{2n}|^{1/n} \rightarrow 0$, then its Legendre series converges locally uniformly in $\mathbb{C}$ and defines an entire function agreeing with f on $[-1, 1]$.*

Proof. For every compact $K \subset \mathbb{C}$ there are constants C_K, R_K with $|P_m(z)| \leq C_K R_K^m$ for $z \in K$. Supergeometric decay makes $\sum a_{2n} P_{2n}(z)$ uniformly convergent on K . The limit is entire and agrees with f on the interval by uniqueness of the Legendre expansion. $\square$

Theorem 9.5 (Rigorous non-flattening of the closed loop; new here). *The central coefficients (73) are unbounded in absolute value:*

$$\limsup_{n \rightarrow \infty} |\beta_n| = \infty. \quad (74)$$

Consequently:

1. the repeated central copies $u(x)$ in (53) cannot be grouped into a scalar coefficient series, because its terms do not tend to zero;
2. the atomwise coefficient variation

$$\Lambda_n = \frac{|u_n|}{4^n} \sum_{|k| < 2 \cdot 4^n} |Q_{2n}(k/4^n)| \quad (75)$$

is unbounded;

3. the double atom expansion is neither absolutely nor unconditionally convergent when flattened term by term, even though its finite blocks form an absolutely and uniformly convergent outer series.

Proof. Suppose $|\beta_n|$ were bounded. By (67),

$$|u_n|^{1/n} \leq \left(\frac{C_0 4^n}{|Q_{2n}(0)|} \right)^{1/n} \rightarrow 0.$$

The even Legendre coefficients of u would therefore be supergeometric. [Theorem 9.4](#) would extend u analytically across the endpoints. But u is C^∞ with compact support, so every derivative at ± 1 vanishes; an analytic extension with a zero Taylor series at an endpoint would be identically zero, contradicting $u(0) = 1$. Hence β_n is unbounded. Since $\Lambda_n \geq |\beta_n|$, the remaining conclusions follow. $\square$

The theorem gives a precise answer to “what happens after closing the loop?” The loop is mathematically exact, but its natural mode of convergence is *blockwise*. Deconvolution creates huge signed coefficients; the Strang–Fix identities then cancel them inside each block to leave the small, smooth function $u_n P_{2n}$.

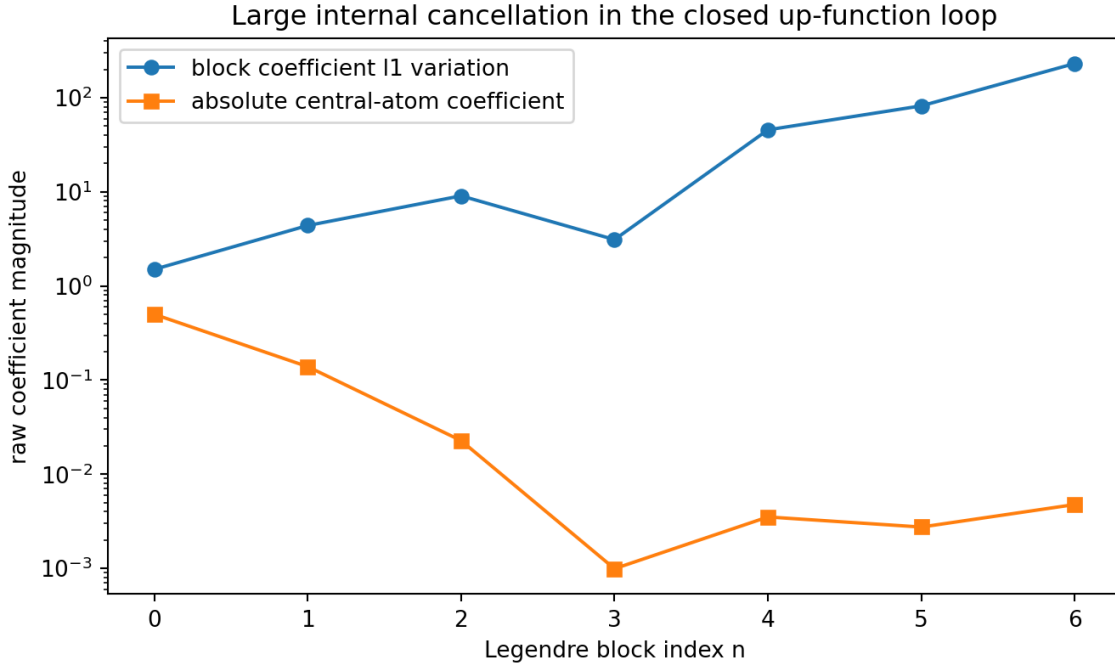


Figure 5: Raw coefficient variation of the first closed-loop blocks. The central coefficient is still small at these low degrees, but [theorem 9.5](#) proves that it is eventually unbounded. The full ℓ^1 variation is already 230.7 at block $n = 6$, although the resulting function is only $u_6 P_{12}$.

10 Connections to q -binomials, Bell polynomials, Thue–Morse, and Lambert W

10.1 A q -geometric row lift of Lagrange weights

The primary corpus proves a reversed Gaussian-binomial formula for Lagrange weights on geometric nodes. The present synthesis lifts that single functional row to an entire up-coefficient matrix.

Take distinct nodes $1, q, \dots, q^d$ over a field and set

$$G_{d+1}(x) = \prod_{r=0}^d (x - q^r) = \sum_{s=0}^{d+1} g_s x^s, \quad g_s = (-1)^{d+1-s} q^{\binom{d+1-s}{2}} \begin{bmatrix} d+1 \\ s \end{bmatrix}_q. \quad (76)$$

Synthetic division gives

$$\frac{G_{d+1}(x)}{x - q^j} = \sum_{n=0}^d h_{n,j} x^n, \quad h_{n,j} = \sum_{s=n+1}^{d+1} g_s q^{j(s-n-1)}. \quad (77)$$

The denominator is

$$G'_{d+1}(q^j) = (-1)^j q^{jd-j(j+1)/2} (q; q)_j (q; q)_{d-j}. \quad (78)$$

Proposition 10.1 (Gaussian-binomial/Appell synthesis matrix; new here). *For geometric nodes q^j and $M = 2^d$, the literal-shift coefficient matrix is*

$$B_{kj}^{(q)} = \frac{1}{M G'_{d+1}(q^j)} \sum_{n=0}^d h_{n,j} A_n(k/M), \quad (79)$$

with $h_{n,j}$ given by (77). Thus every row is a finite convolution of a Gaussian-binomial coefficient row with the Rvachev–Appell family.

Proof. The cardinal polynomial is $\ell_j(x) = G_{d+1}(x)/[(x - q^j)G'_{d+1}(q^j)]$. Insert (77), apply $\mathcal{D}_u x^n = A_n(x)$, and sample as in (13). $\square$

Affine images of geometric nodes can be placed inside $[-1, 1]$. At evaluation $x = 0$, the proposition collapses to the type of q -binomial Lagrange weights already formalized in the corpus; the new content is the simultaneous lift to all dyadic up-shift rows.

10.2 The two different q -products in the loop

Two q -structures coexist but play different roles:

1. geometric interpolation nodes produce Gaussian coefficients $\begin{bmatrix} d+1 \\ s \end{bmatrix}_q$ in (76);
2. dyadic Fourier zero multiplicities produce the annihilator $A_d(z) = \prod_{m=1}^d [2^m]_z$ in (37).

The first is a coordinate choice; the second is intrinsic to the binary sinc product. Their interaction in (79) suggests a two-parameter calculus in which node geometry and dyadic alias geometry can be varied independently.

10.3 Bell–Bernoulli algebra is the coefficient engine

Equations (2), (6), and (12) show that all exact Lagrange-to-up coefficients are finite combinations of:

$$\text{node symmetric functions} \quad \times \quad \text{complete Bell polynomials} \quad \times \quad \text{Bernoulli cumulants}.$$

The same cumulants appear in the nonlocal recurrence (25). Thus Bell and Bernoulli polynomials are not an ornamental reformulation: they are the exact algebraic engine that undoes the smoothing by the random binary uniform series.

10.4 Thue–Morse controls compression, not the literal coefficients

The literal formula (10) follows from Fourier zeros and moment deconvolution. Thue–Morse becomes visible when one asks to remove the exponentially many redundant active atoms on a cell. The corpus identity relating $A_d(z)(z - 1)^{d+1}$ to the length- 2^d Thue–Morse polynomial gives the polynomiality certificate and the $d + 2$ local dimension. In the present matrix language, the cofactor functional λ^T of theorem 6.1 is a normalized coordinate on exactly that same one-dimensional defect quotient.

10.5 Two Lambert geometries

The repository’s W_{-1} endpoint asymptotics invert a small spatial relation of the form $x \asymp \lambda 2^{-\lambda}$. Its reciprocal-MGF coefficient threshold uses the principal branch W to invert factorial growth. The new threshold (72) is a third occurrence: the central binomial coefficient of P_{2n} shifts the scale from $2\pi e$ to πe in the argument of W . The common mechanism is inversion of a quantity $n \log n$, but the constants encode different geometry.

11 Numerical experiments and reproducibility

11.1 Methods

The accompanying script `experiments.py` uses three independent computational layers:

1. exact SymPy arithmetic for cumulants, γ_r , Appell polynomials, Q_n , rational Legendre coefficients, and endpoint-compression matrices;
2. a period-8, 2^{19} -point inverse FFT of the 42-factor sinc product for fast double-precision evaluation of u ;
3. an 80-digit, 3000-term odd-half-integer cosine series for the severely cancellation-prone minimal-dictionary certificate.

No network access is used by the script.

The FFT diagnostics were

$$\int u = 1.0000000000000000, \quad u(0) = 0.9999999999999999, \quad \min u \approx -3.15 \times 10^{-16},$$

where the tiny negative minimum is roundoff.

11.2 Matrix checks through degree ten

d	M	atoms	$\max UB - I $	$\ BW^{-1/2}\ _2$	$\max \text{ column } \ell^1$
1	2	7	1.11×10^{-16}	1.620	2.915
2	4	15	1.23×10^{-11}	2.928	5.537
4	16	63	1.01×10^{-9}	12.437	21.353
6	64	255	3.24×10^{-9}	47.134	91.718
8	256	1023	1.06×10^{-8}	141.924	348.145
10	1024	4095	1.65×10^{-8}	354.264	1212.147

Table 2: Selected rows from `matrix_checks.csv`. The residuals come from FFT interpolation and double-precision cancellation, not from the exact identities.

For every tested degree, the script also checked $UD = V$, $(BU)^2 = BU$, $\text{tr}(BU) = d + 1$, the row-sum identity, the explicit Gauss inverse $V^{-1} = H^{-1}V^TW$, moment transmutation through degree d , and equality of all weighted singular values. The largest residual among the main closure checks was 1.81×10^{-8} .

11.3 Closed-block variation and asymptotic verification

n	u_n	$Q_{2n}(0)$	$ \beta_n $	Λ_n
1	−0.833333	−0.666667	1.389×10^{-1}	4.384
2	0.366667	0.992593	2.275×10^{-2}	9.057
3	0.0252205	−2.51499	9.911×10^{-4}	3.105
4	−0.0857260	10.4961	3.515×10^{-3}	45.549
5	0.0398799	−70.4945	2.745×10^{-3}	81.852
6	−0.0267860	728.295	4.763×10^{-3}	230.706

Table 3: First block coefficients. Low-degree central terms do not yet reveal the eventual unbounded behavior proved in [theorem 9.5](#); the total coefficient variation already grows rapidly.

A separate 100-digit recurrence evaluated (64) up to $n = 50$. At $n = 50$,

$$Q_{100}(0) \approx 4.04295040356 \times 10^{107},$$

the ratio to the leading pole asymptotic in (65) is 1.10433353399, and the root ratio

$$\frac{|Q_{100}(0)|^{1/100}}{100/(\pi e)} = 1.01744592972.$$

The slow approach of the unscaled ratio is consistent with the explicit $\pi^2/(2n)$ first correction, while the root asymptotic converges much faster.

11.4 Reproduction commands and generated files

From the artifact directory, run

```
python experiments.py
```

or omit plots with

```
python experiments.py --no-plots
```

The script regenerates the CSV tables, exact TeX snippets, five diagnostic figures, the high-precision endpoint certificate, and the asymptotic tables. The accompanying `README.txt` records the full inventory.

12 Frontier conjectures and research directions

The exact loop and the non-flattening theorem shift the main questions from existence to geometry, conditioning, and summability.

Conjecture 12.1 (Asymptotic singular-vector localization). *For even degree $d = 2n$, after Gauss weighting, the dominant right singular vector of $D_d H^{-1/2}$ concentrates on the highest few even Legendre modes, and its dominant row mass eventually concentrates on a shrinking neighborhood of the central lattice point $k = 0$. More precisely, there should be $\delta_n \downarrow 0$ such that the fraction of the top left singular vector’s ℓ^2 mass on $|k|/4^n \leq \delta_n$ tends to one.*

The factorial lower bound from $Q_{2n}(0)$ supports the conjecture, but the large finite-degree values near $|k|/M \approx 2$ compete for a long pre-asymptotic range.

Problem 12.2 (Optimal oversampling versus coefficient growth). *The exact reproduction order only requires $\nu_2(m) \geq d$, so one may replace $M = 2^d$ by $m = 2^d q$ with odd q , or alter the physical radius and spacing while preserving their ratio. Determine the choice minimizing a natural stability functional such as*

$$\left\| B_m W^{-1/2} \right\|_2, \quad \max_j \sum_k |B_{m,kj}|, \quad \text{or} \quad \sup_x \sum_k |B_{m,kj} u((x - kh)/\rho)|.$$

The atom-minimal dictionary is not expected to be condition-minimal.

Conjecture 12.3 (Interlacing of the unique nodal-defect spline). *For Gauss–Legendre or Chebyshev–Lobatto nodes, the unique defect function G_Ξ of theorem 6.2 has no zeros other than the prescribed $d + 1$ simple nodal zeros in $(-1, 1)$, with alternating sign on consecutive node intervals.*

A proof would amount to an extended-Chebyshev property for the minimal endpoint up-dictionary and would make the rank-one complement geometrically canonical, not merely algebraically canonical.

Conjecture 12.4 (Sign regularity for geometric-node synthesis). *For $0 < q < 1$, after the natural checkerboard row and column sign changes, the matrix $B^{(q)}$ in (79) is sign regular of order $d + 1$ on the active dyadic lattice. Its maximal minors should admit product formulae combining Gaussian coefficients with the dyadic annihilator.*

This would connect total positivity of interpolation systems to the q-binomial/Appell row lift and could lead to stable barycentric variants.

Problem 12.5 (Borel and resurgent interpretation of the formal Q generating function). *Develop a Borel–Laplace summation theory for (71). The reciprocal-MGF pole shells at $\pm 2\pi i n$, with multiplicity $1 + \nu_2(n)$, should reappear as a 2-adically stratified singularity set in the Borel plane. Determine whether a canonical lateral sum exists away from the corresponding rays and whether its Stokes jumps have a direct Thue–Morse interpretation.*

Problem 12.6 (Preconditioned exact loops). *Find a polynomial basis or a biorthogonal up-atom dictionary in which the Legendre partial loop remains exact but the coefficient blocks have controlled variation. The weighted identity (33) shows that merely switching between Gauss Lagrange and Legendre coordinates cannot solve the problem; a successful preconditioner must alter the deconvolution stage or the atom dictionary itself.*

12.1 Promising computational directions

Several concrete experiments appear particularly informative:

- compute the full singular spectrum of $D_d H^{-1/2}$ in arbitrary precision for d up to at least 40, comparing central and edge lattice localization;
- track zeros of Q_n and of the nodal-defect spline G_Ξ as d grows;
- optimize odd overscaling factors q for stability, not just exact order;
- search for closed q-hypergeometric simplifications of (79);
- compare Gauss, Chebyshev, Leja, and geometric nodes through the exact projector norm $\|BU\|$ and the certificate vector λ ;
- use interval arithmetic to certify the high-degree asymptotic and singular-value experiments.

13 Formalization roadmap

The results are well suited to a modular Lean development because the analytic input has largely already been formalized in the repository.

1. **Polynomial deconvolution API.** Define $\mathcal{D}_u$ on `Polynomial` as the terminating reciprocal-MGF differential operator; prove linearity, degree preservation, parity, and $M_u(D)\mathcal{D}_u p = p$.
2. **Lagrange specialization.** Instantiate the existing interval synthesis theorem with $[-1, 1]$, $h = 2^{-d}$, and $\rho = 1$; prove the two endpoint atoms vanish and obtain (10).
3. **Finite matrices.** Use `Matrix` indexed by `Fin (d+1)` and the finite integer interval $|k| < 2^{d+1}$; prove $UB = 1$, then the projector and moment identities.
4. **Basis triangle.** Package $UD = V$, $B = DV^{-1}$, and specialize to Legendre polynomials.
5. **Gauss kernel.** Import or prove finite Gauss quadrature orthogonality, $V^T WV = H$, the cardinal kernel identity, and the weighted singular-value factorization.
6. **Minimal cofactor theorem.** Formalize alternating maximal minors for a rectangular matrix of corank one and derive $I - \widehat{B}\widehat{U} = v\lambda^T$ from Cauchy–Binet.
7. **Closed loop.** Combine the formal Fourier–Legendre series with finite block identities. Keep the outer series in a block-valued Banach space to make the legal grouping explicit.
8. **Asymptotic and non-flattening.** Reuse the formalized or semi-formalized reciprocal coefficient asymptotic, prove the finite Legendre-coefficient estimate, and formalize the analytic-extension contradiction for supergeometric Legendre coefficients.

The matrix and cofactor parts are finite algebra. The only genuinely new analytic bridge needed for the final theorem is a convenient formal criterion that supergeometric Legendre coefficients imply analytic extension.

14 Conclusion

The requested finite representation exists in a particularly explicit form: for any nodes on $[-1, 1]$, every Lagrange cardinal polynomial is obtained by sampling one deconvolved polynomial on the dyadic lattice and attaching those samples to literal translates of u . This turns interpolation into an exact right-inverse problem, not an approximate radial-basis fit.

The Gauss–Legendre specialization closes the conceptual gap between Lagrange and Legendre. Its deconvolved Christoffel–Darboux kernel gives a compact formula for the entire coefficient matrix, while the weighted singular-value identity shows precisely where instability enters. The endpoint compression reveals the one-dimensional quotient beyond polynomials, and cofactors provide a canonical rank-one defect decomposition.

Substituting the finite Legendre synthesis into the repository’s canonical series does close the loop, both termwise and at each fixed partial-sum scale. But the loop is not a benign self-expansion. The new asymptotic for $Q_{2n}(0)$ and the nonanalyticity of u force the repeated central-atom coefficients to be unbounded. The correct mathematical object is therefore an absolutely convergent series of exact finite cancellation blocks. Any future stable algorithm must respect that block geometry or redesign the dictionary; it cannot simply flatten the atoms and hope that ordinary absolute convergence survives.

A Exact low-degree and matrix data

The files `symbolic_data.txt`, `low_degree_polynomials.tex`, and `legendre_coefficients.tex` in the archive contain exact values through the degrees used by the experiments. For reference, the reciprocal coefficients begin

$$\begin{aligned} \gamma_0 &= 1, & \gamma_2 &= -\frac{1}{9}, & \gamma_4 &= \frac{31}{675}, & \gamma_6 &= -\frac{2347}{59535}, \\ \gamma_8 &= \frac{1904369}{32531625}, & \gamma_{10} &= -\frac{3309760579}{24405225075}. \end{aligned}$$

They satisfy the Bell recurrence

$$\gamma_n = - \sum_{r=1}^n \binom{n-1}{r-1} \kappa_r \gamma_{n-r}. \quad (80)$$

B Endpoint-compression algorithm in explicit form

For completeness, this appendix records the exact specialization used by the code. Let $P(t) = p(2t-1)$, $M = 2^d$, and

$$Q(t) = M_u(MD_t)^{-1}P(t).$$

The active integer centers on $[0, 1]$ are $-M+1, -M+2, \dots, M$. Write

$$A_d(z) = \sum_{r=0}^{R_d} a_r z^r, \quad R_d = 2M - d - 2, \quad a_{R_d} = 1.$$

Initialize a null sequence by

$$n_k = Q(k), \quad -M+1 \leq k \leq -M+R_d,$$

and extend it by the monic recurrence

$$n_{s+R_d} = - \sum_{r=0}^{R_d-1} a_r n_{s+r}. \quad (81)$$

Then

$$b_k = Q(k) - n_k, \quad M-d-1 \leq k \leq M,$$

are the unique endpoint coefficients and give

$$p(x) = \frac{1}{M} \sum_{k=M-d-1}^M b_k u\left(\frac{x+1-2k}{2M}\right).$$

Applying this columnwise to the cardinals produces $\hat{B}$.

C Artifact inventory

The ZIP archive accompanying this report contains:

- `lagrange_rvachev_loop.tex` and the compiled PDF;
- `experiments.py`, fully commented and executable offline;
- `README.txt` with reproduction instructions;
- CSV diagnostics: `matrix_checks.csv` and `interpolation_errors.csv`;
`block_instability.csv` and `q_origin_asymptotics.csv`;
- `endpoint_compression_certificate.txt` and exact TeX data snippets;
- the PNG figures used in the report.

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